1. Understand the Dataset

In [15]:
```
# 1.1 Import the dataset
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

In [16]:
```
df = pd.read_csv('customer_analysis.csv')
```

In [17]:
```
# 1.2 Visualize the dataset
df.head()
```

Out[17]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk

5 rows × 53 columns

In [29]:
```
# 1.3 Print columns
df.columns
```

Out[29]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location',
       'Request_Closing_Time'],
      dtype='str')
```

In [18]:
```
# 1.4 Identify the shape of the dataset
df.shape
```

Out[18]:
```
(364558, 53)
```

In [19]:
```
df.duplicated().sum()
```

Out[19]:
```
np.int64(0)
```

In [20]:
```
# 1.5 Identify the variables with null values
df.isnull().info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   Unique Key                      364558 non-null  bool
 1   Created Date                    364558 non-null  bool
 2   Closed Date                     364558 non-null  bool
 3   Agency                          364558 non-null  bool
 4   Agency Name                     364558 non-null  bool
 5   Complaint Type                  364558 non-null  bool
 6   Descriptor                      364558 non-null  bool
 7   Location Type                   364558 non-null  bool
 8   Incident Zip                    364558 non-null  bool
 9   Incident Address                364558 non-null  bool
 10  Street Name                     364558 non-null  bool
 11  Cross Street 1                  364558 non-null  bool
 12  Cross Street 2                  364558 non-null  bool
 13  Intersection Street 1           364558 non-null  bool
 14  Intersection Street 2           364558 non-null  bool
 15  Address Type                    364558 non-null  bool
 16  City                            364558 non-null  bool
 17  Landmark                        364558 non-null  bool
 18  Facility Type                   364558 non-null  bool
 19  Status                          364558 non-null  bool
 20  Due Date                        364558 non-null  bool
 21  Resolution Description          364558 non-null  bool
 22  Resolution Action Updated Date  364558 non-null  bool
 23  Community Board                 364558 non-null  bool
 24  Borough                         364558 non-null  bool
 25  X Coordinate (State Plane)      364558 non-null  bool
 26  Y Coordinate (State Plane)      364558 non-null  bool
 27  Park Facility Name              364558 non-null  bool
 28  Park Borough                    364558 non-null  bool
 29  School Name                     364558 non-null  bool
 30  School Number                   364558 non-null  bool
 31  School Region                   364558 non-null  bool
 32  School Code                     364558 non-null  bool
 33  School Phone Number             364558 non-null  bool
 34  School Address                  364558 non-null  bool
 35  School City                     364558 non-null  bool
 36  School State                    364558 non-null  bool
 37  School Zip                      364558 non-null  bool
 38  School Not Found                364558 non-null  bool
 39  School or Citywide Complaint    364558 non-null  bool
 40  Vehicle Type                    364558 non-null  bool
 41  Taxi Company Borough            364558 non-null  bool
 42  Taxi Pick Up Location           364558 non-null  bool
 43  Bridge Highway Name             364558 non-null  bool
 44  Bridge Highway Direction        364558 non-null  bool
 45  Road Ramp                       364558 non-null  bool
 46  Bridge Highway Segment          364558 non-null  bool
 47  Garage Lot Name                 364558 non-null  bool
 48  Ferry Direction                 364558 non-null  bool
 49  Ferry Terminal Name             364558 non-null  bool
 50  Latitude                        364558 non-null  bool
 51  Longitude                       364558 non-null  bool
 52  Location                        364558 non-null  bool
dtypes: bool(53)
memory usage: 18.4 MB
```

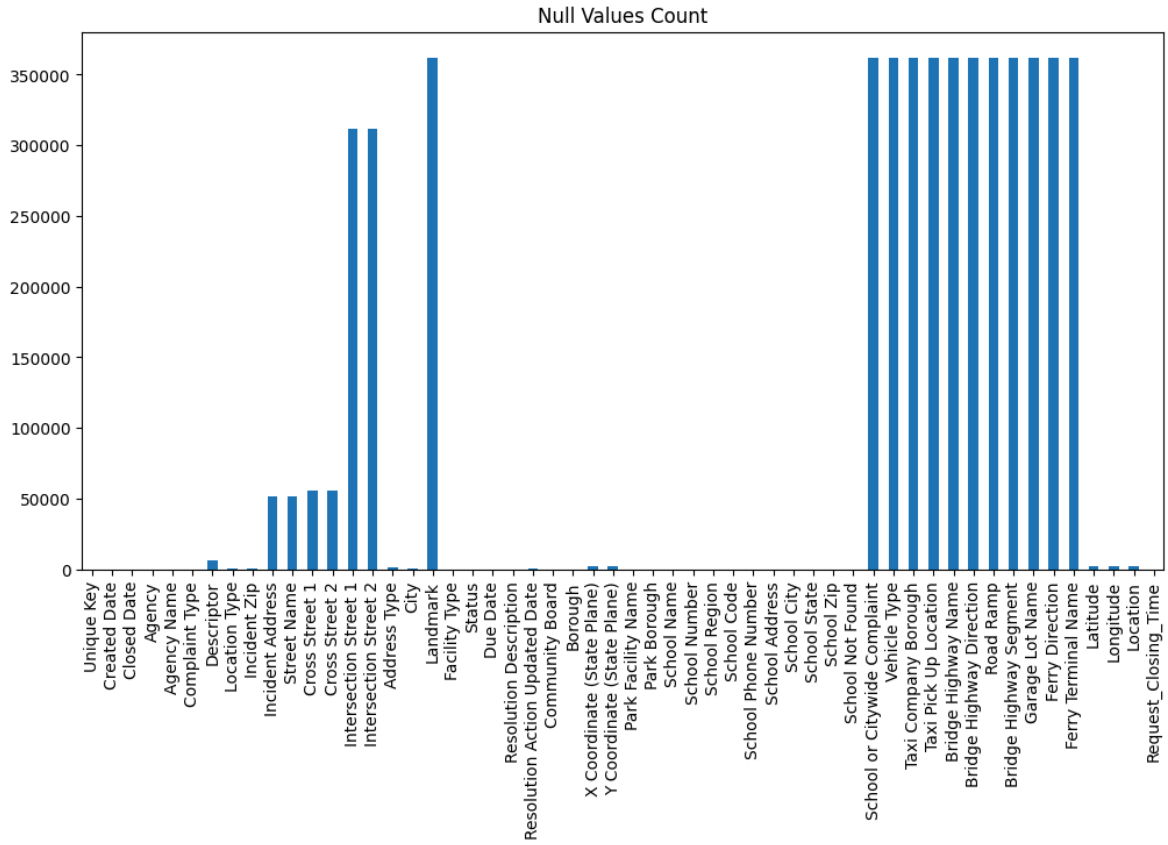
In [21]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   Unique Key                      364558 non-null  int64
 1   Created Date                    364558 non-null  str
 2   Closed Date                     362177 non-null  str
 3   Agency                          364558 non-null  str
 4   Agency Name                     364558 non-null  str
 5   Complaint Type                  364558 non-null  str
 6   Descriptor                      358057 non-null  str
 7   Location Type                   364425 non-null  str
 8   Incident Zip                    361560 non-null  float64
 9   Incident Address                312859 non-null  str
 10  Street Name                     312859 non-null  str
 11  Cross Street 1                  307370 non-null  str
 12  Cross Street 2                  306753 non-null  str
 13  Intersection Street 1           51120 non-null   str
 14  Intersection Street 2           50512 non-null   str
 15  Address Type                    361306 non-null  str
 16  City                            361561 non-null  str
 17  Landmark                        375 non-null     str
 18  Facility Type                   362169 non-null  str
 19  Status                          364558 non-null  str
 20  Due Date                        364555 non-null  str
 21  Resolution Description          364558 non-null  str
 22  Resolution Action Updated Date  362156 non-null  str
 23  Community Board                 364558 non-null  str
 24  Borough                         364558 non-null  str
 25  X Coordinate (State Plane)      360528 non-null  float64
 26  Y Coordinate (State Plane)      360528 non-null  float64
 27  Park Facility Name              364558 non-null  str
 28  Park Borough                    364558 non-null  str
 29  School Name                     364558 non-null  str
 30  School Number                   364558 non-null  str
 31  School Region                   364557 non-null  str
 32  School Code                     364557 non-null  str
 33  School Phone Number             364558 non-null  str
 34  School Address                  364558 non-null  str
 35  School City                     364558 non-null  str
 36  School State                    364558 non-null  str
 37  School Zip                      364557 non-null  str
 38  School Not Found                364558 non-null  str
 39  School or Citywide Complaint    0 non-null       float64
 40  Vehicle Type                    0 non-null       float64
 41  Taxi Company Borough            0 non-null       float64
 42  Taxi Pick Up Location           0 non-null       float64
 43  Bridge Highway Name             297 non-null     str
 44  Bridge Highway Direction        297 non-null     str
 45  Road Ramp                       262 non-null     str
 46  Bridge Highway Segment          262 non-null     str
 47  Garage Lot Name                 0 non-null       float64
 48  Ferry Direction                 1 non-null       str
 49  Ferry Terminal Name             2 non-null       str
 50  Latitude                        360528 non-null  float64
 51  Longitude                       360528 non-null  float64
 52  Location                        360528 non-null  str
dtypes: float64(10), int64(1), str(42)
memory usage: 333.0 MB
```

2.Exploratory Data Analysis (EDA)

In [34]:
```
# 2.1Draw a frequency plot to show the number of null values in each column of t
df.isnull().sum().plot(kind='bar', figsize=(12,6))

plt.title("Null Values Count")
plt.xticks(rotation=90)   # rotate x-axis labels
plt.show()
```



In [35]:
```
# 2.2 Missing value treatment
# 2.2.1 Remove records where "Closed Date" is null
df = df.dropna(subset=['Closed Date'])
```

In [36]:
```
# 2.3 Convert "Created Date" and "Closed Date" to datetime format
df['Created Date'] = pd.to_datetime(df['Created Date'])
df['Closed Date'] = pd.to_datetime(df['Closed Date'])
df.head()
```

Out[36]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Locat
0	32310363	2015-12-31 23:59:45	2016-01-01 00:55:15	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/
1	32309934	2015-12-31 23:59:44	2016-01-01 01:26:57	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/
2	32309159	2015-12-31 23:59:29	2016-01-01 04:51:03	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/
3	32305098	2015-12-31 23:57:46	2016-01-01 07:43:13	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Street/
4	32306529	2015-12-31 23:56:58	2016-01-01 03:24:42	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Street/

5 rows × 54 columns

In [37]:
```
# 2.3.1Create a new column 'Request_Closing_Time' as the difference between Clos
df['Request_Closing_Time'] = df ['Closed Date'] - df['Created Date']
df['Request_Closing_Time']
```

Out[37]:
```
0        0 days 00:55:30
1        0 days 01:27:13
2        0 days 04:51:34
3        0 days 07:45:27
4        0 days 03:27:44
                ...
364553   0 days 10:17:47
364554   0 days 02:20:34
364555   0 days 00:19:03
364556   0 days 02:40:53
364557   0 days 02:47:00
Name: Request_Closing_Time, Length: 362177, dtype: timedelta64[us]
```

In [39]:
```
# 2.3.2Convert the calculated date to seconds to get a better representation
df['Request_Closing_Time'] = df['Request_Closing_Time'].dt.total_seconds()
```

In [43]:
```
# 2.3.3 View the descriptive statistics for the newly created column
df['Request_Closing_Time'].describe()
```

Out[43]:
```
count    3.621770e+05
mean     1.511330e+04
std      2.110255e+04
min      6.100000e+01
25%      4.533000e+03
50%      9.616000e+03
75%      1.887800e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

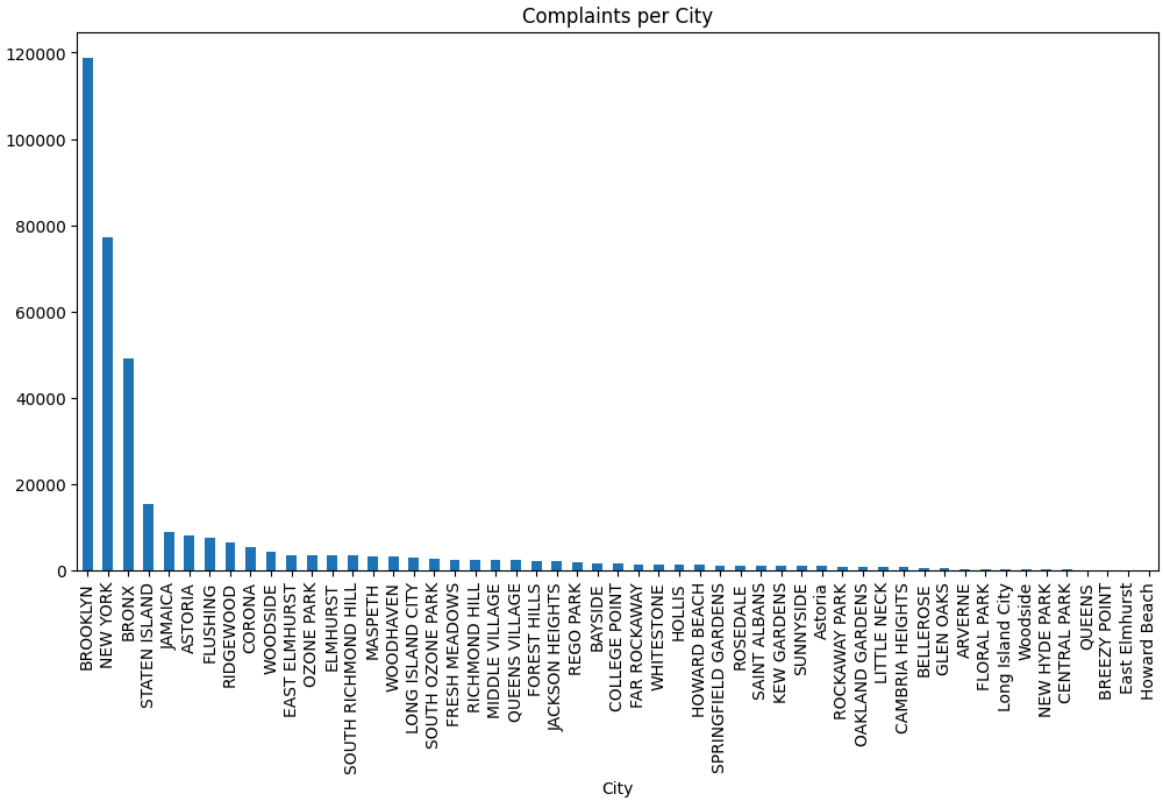
In [44]:
```
# 2.3.4Check the number of null values in the Complaint_Type and City columns
df[['Complaint Type', 'City']].isnull().sum()
```

Out[44]:
```
Complaint Type      0
City              674
dtype: int64
```

In [46]:
```
# 2.3.5Impute the NA value with Unknown City
df['City'].fillna("Unknown City", inplace=True)
df['City']
```

Out[46]:
```
0                 NEW YORK
1                  ASTORIA
2                    BRONX
3                    BRONX
4                 ELMHURST
                 ...
364553           WOODHAVEN
364554               BRONX
364555            NEW YORK
364556               BRONX
364557    SOUTH OZONE PARK
Name: City, Length: 362177, dtype: str
```

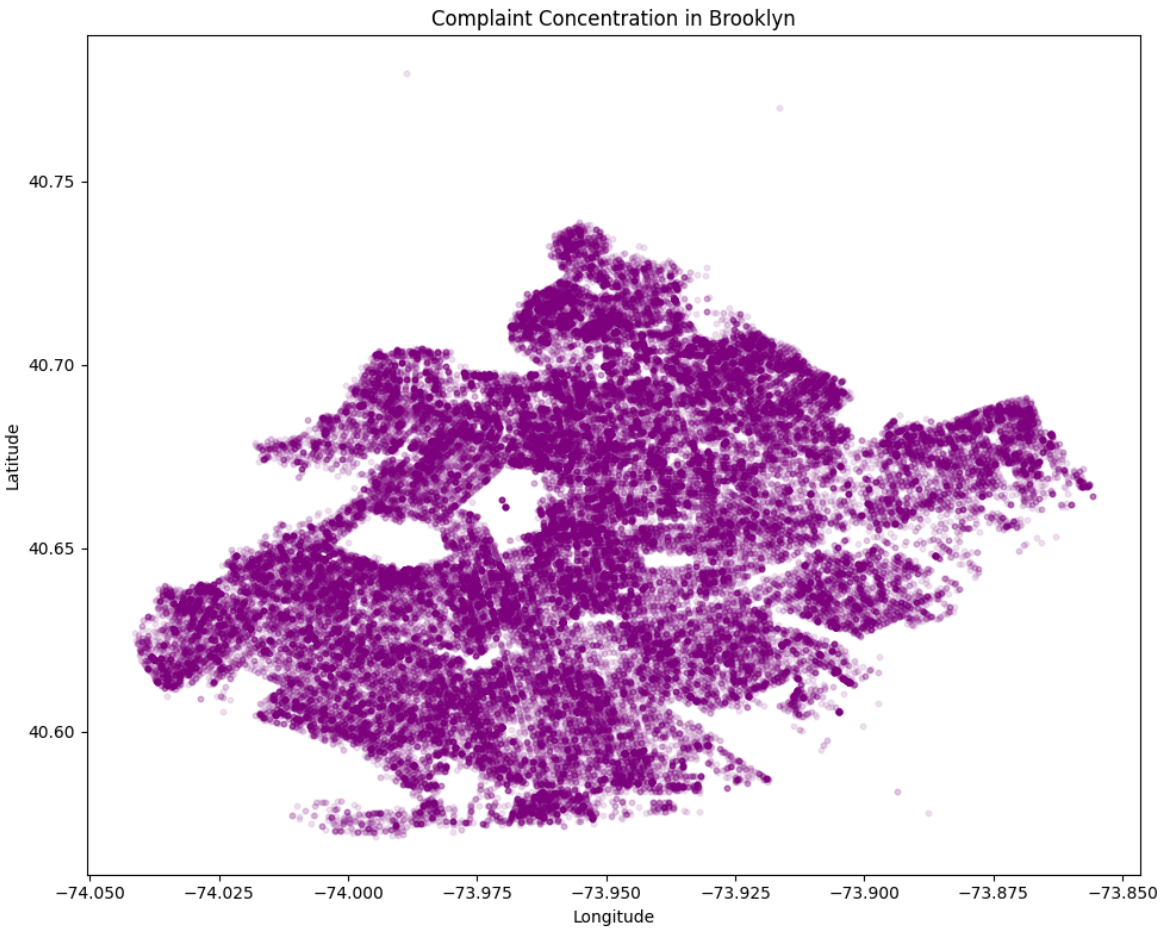
In [48]:
```
# 2.3.6 Draw a frequency plot for the complaints in each city
df['City'].value_counts().plot(kind='bar', figsize=(12,6))
plt.title("Complaints per City")
plt.show()
```



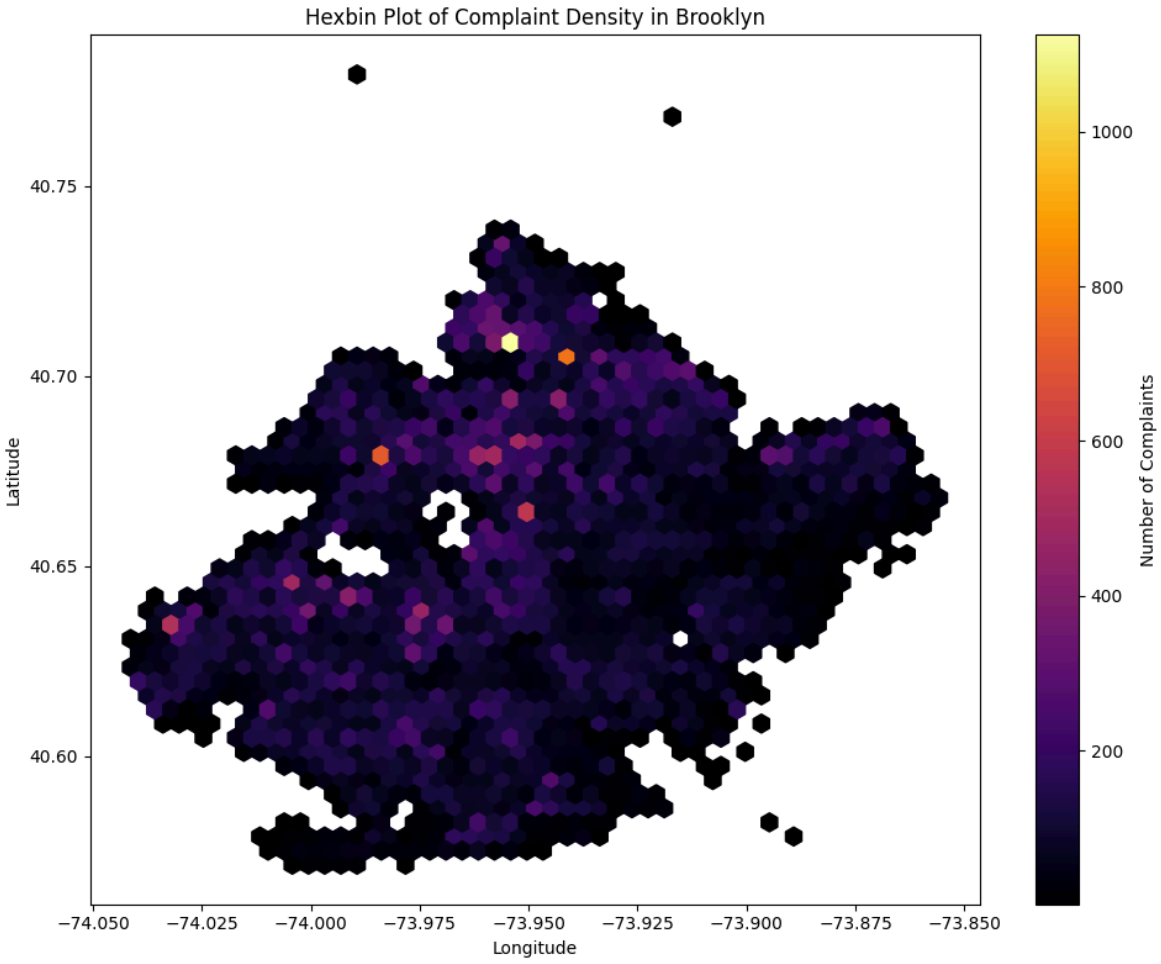
In [69]:
```
# 2.3.7Create a scatter and hexbin plot of the concentration of complaints acros
brooklyn_data = df[df['City'].str.upper() == 'BROOKLYN'].dropna(subset=['Latitud

# Scatter plot for complaint concentration in Brooklyn (using latitude & longitu
plt.figure(figsize=(10, 8))
plt.scatter(brooklyn_data['Longitude'], brooklyn_data['Latitude'], alpha=0.1, s=
```

```
plt.title('Complaint Concentration in Brooklyn')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.tight_layout()
plt.show()
```

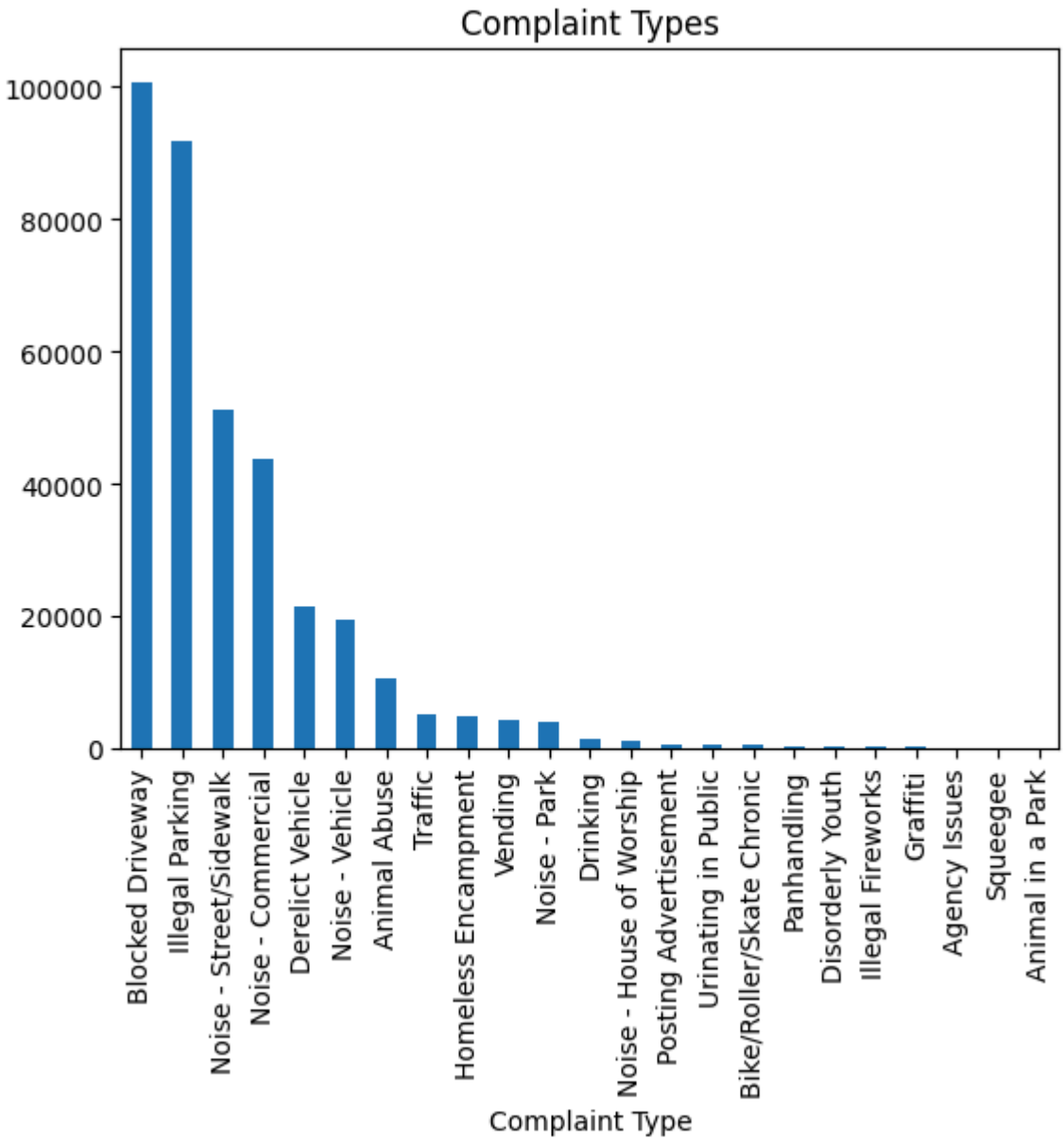


In [70]:
```
# Hexbin plot for density visualization (Optional but recommended)
plt.figure(figsize=(10, 8))
plt.hexbin(brooklyn_data['Longitude'], brooklyn_data['Latitude'], gridsize=50, c
plt.colorbar(label='Number of Complaints')
plt.title('Hexbin Plot of Complaint Density in Brooklyn')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.tight_layout()
plt.show()
```



3.Major Types of Complaints

In [51]:
```
# 3.1Plot a bar graph to show the types of complaints
df['Complaint Type'].value_counts().plot(kind = 'bar')
plt.title("Complaint Types")
plt.show()
```



In [53]:
```
# 3.2Check the frequency of various types of complaints for New York City
nyc = df[df['City'] == 'NEW YORK']
nyc['Complaint Type'].value_counts()
```

Out[53]:
```
Complaint Type
Noise - Street/Sidewalk      22245
Noise - Commercial           18686
Illegal Parking              14549
Noise - Vehicle               6294
Homeless Encampment           3060
Blocked Driveway              2705
Vending                       2638
Animal Abuse                  1941
Traffic                       1769
Noise - Park                  1243
Derelict Vehicle               695
Drinking                       321
Urinating in Public            264
Bike/Roller/Skate Chronic      254
Noise - House of Worship       222
Panhandling                    206
Disorderly Youth                81
Posting Advertisement           49
Illegal Fireworks               38
Graffiti                        25
Squeegee                         4
Name: count, dtype: int64
```

In [79]:
```
# 3.3 Find the top 10 complaint types
df['Complaint Type'].value_counts().head(10)
```

Out[79]:
```
Complaint Type
Blocked Driveway           100624
Illegal Parking             91716
Noise - Street/Sidewalk     51139
Noise - Commercial          43751
Derelict Vehicle            21518
Noise - Vehicle             19301
Animal Abuse                10530
Traffic                      5196
Homeless Encampment          4879
Vending                      4185
Name: count, dtype: int64
```

In [55]:
```
# 3.4 Display the various types of complaints in each city
df.groupby('City')['Complaint Type'].value_counts()
```

Out[55]:
```
City      Complaint Type
ARVERNE   Illegal Parking            62
          Blocked Driveway           50
          Animal Abuse               46
          Derelict Vehicle           32
          Noise - Street/Sidewalk    29
                                   ...
Woodside  Illegal Parking           124
          Blocked Driveway           27
          Derelict Vehicle            8
          Noise - Street/Sidewalk     5
          Noise - Commercial          2
Name: count, Length: 777, dtype: int64
```

In [60]:
```
# 3.5Create a DataFrame, df_new, which contains cities as columns and complaint
df_new = pd.crosstab(df['Complaint Type'], df['City'])
```

```
df_new.head()
```

Out[60]:

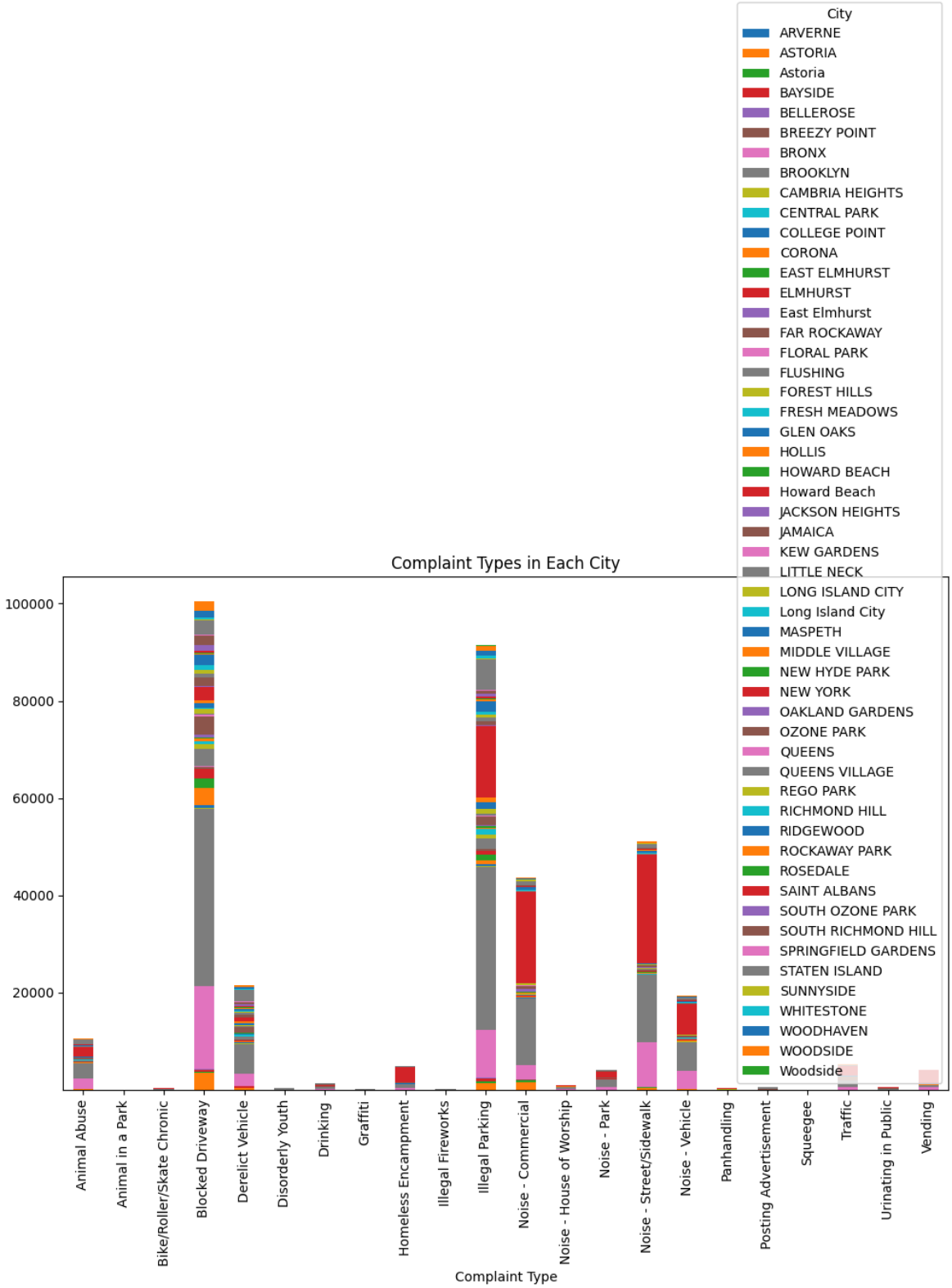
City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX
Complaint Type							
Animal Abuse	46	170	0	53	15	2	1971
Animal in a Park	0	0	0	0	0	0	0
Bike/Roller/Skate Chronic	0	16	0	0	1	0	22
Blocked Driveway	50	3436	159	514	138	3	17062
Derelict Vehicle	32	426	14	231	120	3	2402

5 rows × 53 columns

4.Visualize the major types of complaints in each city

In [63]:

```
# 4.1Draw another chart that shows the types of complaints in each city in a sin
# where different colors show the different types of complaints
df_new.plot(kind ='bar', stacked = True, figsize = (12,7))
plt.title("Complaint Types in Each City")
plt.show()
```

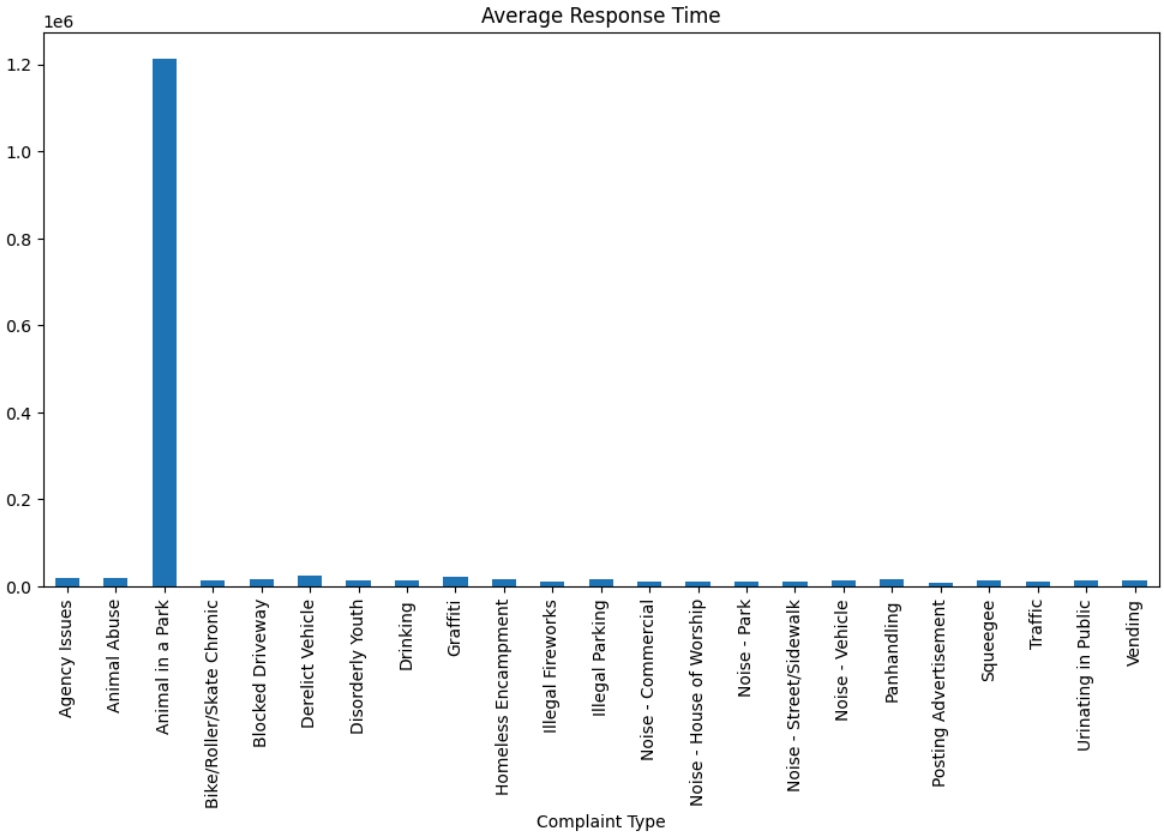


In [64]:
```
# 4.2Sort the complaint types based on the average Request_Closing_Timegrouping
#different locations
avg_time = df.groupby(['City', 'Complaint Type'])['Request_Closing_Time'].mean()
avg_time.sort_values().head(10)
```

Out[64]:
```
City                 Complaint Type
ARVERNE              Drinking                      859.0
RIDGEWOOD            Posting Advertisement         895.0
EAST ELMHURST        Bike/Roller/Skate Chronic     908.0
OZONE PARK           Illegal Fireworks            1191.0
QUEENS               Urinating in Public          1226.0
SOUTH RICHMOND HILL  Illegal Fireworks            1712.5
ARVERNE              Vending                      1735.0
SOUTH OZONE PARK     Illegal Fireworks            1821.0
KEW GARDENS          Posting Advertisement        1872.0
SUNNYSIDE            Graffiti                     2133.0
Name: Request_Closing_Time, dtype: float64
```

5.Average Response Time

In [66]:
```
# 5.1Visualize the average of Request_Closing_Time
avg = df.groupby('Complaint Type')['Request_Closing_Time'].mean()
avg.plot(kind='bar', figsize=(12,6))
plt.title("Average Response Time")
plt.show()
```



6. Identify the significant variables by performing statistical analysis using p-values

7.Perform a Kruskal-WallisH test

In [82]:
```
#7.1 Fail to reject H0: All sample distributions are equal
#7.2 Reject H0: One or more sample distributions are not equal
```

```python
from scipy.stats import kruskal

groups = df.groupby('Complaint Type')['Request_Closing_Time'].apply(list)

stat, p = kruskal(*groups[:5])  # using top 5 groups

print("P-value:", p)

if p < 0.05:
    print("Reject H0 → Not equal")
else:
    print("Fail to Reject H0 → Equal")
```

```
P-value: 2.0456061161139882e-13
Reject H0 → Not equal
```

8. 📊 Observations:

📊 Observations:

- Most complaints are concentrated in major cities like New York, Brooklyn, and Queens.
- A few complaint types occur much more frequently than others, indicating common recurring issues.
- Missing values were mainly present in location-related columns and were handled properly.
- Response time varies significantly across different complaint types.
- Brooklyn shows a high density of complaints in specific geographic areas.
- The top 10 complaint types dominate the overall dataset distribution.
- Different cities show different patterns of complaint types.
- The average response time is not uniform across all complaint categories.
- The Kruskal-Wallis test indicates that response time is not the same for all complaint types (when p-value < 0.05).